HEAT-FLOW PDE HIERARCHY ON JENSEN INVARIANTS, NODAL-CURVATURE ZERO VELOCITY, AND CONDITIONAL SADDLE PERTURBATION UNDER DE BRUIJN–NEWMAN EVOLUTION

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Abstract. Let $\Xi_\tau(z) := 8\int_0^\infty \Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u = 8H_\tau(2z)$ be the De Bruijn–Newman deformation of Riemann's completed zeta function, where $\tau = t_{\mathrm{DBN}}$ is the standard De Bruijn–Newman parameter ($\partial_\tau\Xi_\tau = -\frac{1}{4}\partial_z^2\Xi_\tau$) and $W_\tau(d,t) := \frac{1}{2}\partial_d|\Xi_\tau(t-\mathrm{i}d)|^2 = 16\int_0^\infty y\sinh(2dy)I_\tau(y,t)\,\mathrm{d}y$. First, we relate the Csordas–Smith–Varga and Rodgers–Tao velocity $\gamma_k'(0) = \Xi''(\gamma_k)/(4\Xi'(\gamma_k))$ to the parabolic curvature $\kappa(\gamma_k)$ of $R(d,t) = 0$ and the extremum curvature on $I(d,t) = 0$. Second, under $y = u+v$, $x = u-v$, we establish the cross-term-free Gaussian Wigner factorization, exhibit the boundary zero $G_{0,\tau}(\mathrm{i}\eta) = 0$ at $2\eta_0 = 0.63883\cdots < \frac{\pi}{4}$, and isolate the numerator condition for pitchfork saddle perturbation. Third, we prove a second-order radial wave-heat PDE for $\widetilde{W_\tau} = W_\tau/d$ and the heat-flow hierarchy $(\partial_\tau + \frac{1}{8}\partial_t^2)L_m(\Xi_\tau;t) = \frac{(m+1)(2m+1)}{4}L_{m+1}(\Xi_\tau;t)$ on all Jensen invariants, yielding the $(m-1)^2/2$ multiple-zero coalescence lift and conditional Lehmer-pair scaling.

1. Introduction

In the classical theory of the De Bruijn–Newman constant Λ initiated by Pólya [14], de Bruijn [7], and Newman [13], and developed quantitatively by Csordas, Norfolk, Smith, and Varga [3, 4, 5], Rodgers and Tao [17], and the Polymath 15 collaboration [15], one studies the backward-heat family

$$H_\tau(x) := \int_0^\infty \mathrm{e}^{\tau u^2}\,\Phi(u)\cos(xu)\,\mathrm{d}u, \qquad \Phi(u) := \sum_{n=1}^\infty (2\pi^2 n^4\mathrm{e}^{9u} - 3\pi n^2\mathrm{e}^{5u})\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}}, \tag{1.1}$$

satisfying $\partial_\tau H_\tau(x) = -\partial_x^2 H_\tau(x)$, whose zeros are all real if and only if $\tau \geq \Lambda$. The Riemann Hypothesis is equivalent to $\Lambda \leq 0$, while Newman's conjecture $\Lambda \geq 0$ was proved unconditionally by Rodgers and Tao [17], and Polymath 15 [15] established $\Lambda \leq 0.22$.

Throughout this series we work in spectral height $z = t - \mathrm{i}d$ ($s = \frac{1}{2} + d + \mathrm{i}t$), related to (1.1) by $x = 2z$:

$$\Xi_\tau(z) := 8\int_0^\infty \Phi_\tau(u)\cos(2zu)\,\mathrm{d}u = 8\,H_\tau(2z), \qquad \Phi_\tau(u) := \Phi(u)\,\mathrm{e}^{\tau u^2}. \tag{1.2}$$

Because the Gaussian factor $\mathrm{e}^{\tau u^2}$ in (1.2) uses the *identical* variable u as (1.1), the heat parameter τ is identically the standard De Bruijn–Newman time:

$$\tau = t_{\mathrm{DBN}}, \qquad \frac{\partial\Xi_\tau}{\partial\tau}(z) = -\frac{1}{4}\frac{\partial^2\Xi_\tau}{\partial z^2}(z), \tag{1.3}$$

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where the prefactor $-\frac{1}{4}$ arises solely from the spatial rescaling $x = 2z$ ($\partial_z^2 = 4\partial_x^2$), and the De Bruijn–Newman threshold in τ is Λ itself (not 4Λ). At $\tau = 0$, $\Xi_0(z) = \Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$ [16, 19]. In canonical normalization, the deformed transverse Wronskian and its desingularized quotient $\widetilde{W}_\tau(d,t) := W_\tau(d,t)/d$ read

$$W_\tau(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\left|\Xi_\tau(t - \mathrm{i}d)\right|^2 = 16\int_0^\infty y\sinh(2dy)\, I_\tau(y,t)\,\mathrm{d}y, \tag{1.4}$$

where $I_\tau(y,t) := \int_{-\infty}^{\infty} \Phi_\tau(\frac{y+x}{2})\Phi_\tau(\frac{y-x}{2})\cos(2tx)\,\mathrm{d}x$ is the Wigner-type autocorrelation [20, 9]. By Speiser [18], de Branges [6], Hinkkanen [8], and Lagarias [11], Ξ_τ is zero-free in $d > 0$ if and only if $W_\tau(d,t) > 0$ on $d > 0$.

1.1. **Proof overview.** This paper clarifies and proves three connections between the longitudinal evolution (1.3) and the transverse strip geometry $d \geq 0$, separating unconditional differential identities from domain and sign hypotheses:

(1) **Geometric interpretation of the Csordas–Smith–Varga / Rodgers–Tao velocity (Section 2).** For simple zeros $x_k(\tau) = 2\gamma_k(\tau)$ of $H_\tau(x)$, Csordas, Smith, and Varga [5] and Rodgers and Tao [17] proved the Dyson ODE $x_k'(\tau) = H_\tau''(x_k)/H_\tau'(x_k) = 2\,\mathrm{P.V.}\sum_{j\neq k}(x_k - x_j)^{-1}$, so that $\gamma_k'(0) = \Xi''(\gamma_k)/(4\Xi'(\gamma_k))$ equals $\frac{1}{2}\,\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}(\gamma_k - \gamma_j)^{-1}$. Proposition 2.1 records the harmonic X-ray interpretation [18, 1]: because $\partial_d^2 R = -\partial_t^2 R$ and $\partial_\tau \Xi_\tau = -\frac{1}{4}\partial_t^2\Xi_\tau$ share the operator $-\partial_t^2$, $\gamma_k'(0) = \frac{1}{4}\kappa(\gamma_k)$ equals one-fourth the parabolic curvature $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0)$ of $R(d,t) = 0$ at $d = 0$, while critical-line extrema $\tau_k(\tau)$ ($\partial_t\Xi_\tau(\tau_k) = 0$) satisfy $\tau_k'(0) = \frac{3}{4}\kappa_\Gamma(\tau_k)$ relative to $I(d,t)/d = 0$.

(2) **Mirror Gaussian factorization, imaginary-axis kernel zero, and conditional saddle shift (Section 3).** Under $y = u + v$, $x = u - v$, the parallelogram identity $u^2 + v^2 = \frac{1}{2}(y^2 + x^2)$ factors $G_{y,\tau}(x) := \Phi_\tau(\frac{y+x}{2})\Phi_\tau(\frac{y-x}{2}) = \mathrm{e}^{\frac{\tau}{2}y^2}\mathrm{e}^{\frac{\tau}{2}x^2}G_{y,0}(x)$ without an xy cross-term (Proposition 3.1). Crucially, however, $G_{y,\tau}(\mathrm{i}\eta)$ is *not* globally positive on $[0,\infty)\times(-\frac{\pi}{4},\frac{\pi}{4})$: at $y = 0$, $\Phi(\mathrm{i}\eta/2)$ vanishes at $2\eta_0 = 0.6388300536\cdots < \frac{\pi}{4}$. Restricting to a zero-free neighborhood of a non-degenerate pitchfork saddle (y_c,η_c), Proposition 3.2 derives $\partial_\tau y_c(\omega;0)$ and $\partial_\tau\eta_c(\omega;0)$, isolating the numerator inequality $1 - (\partial_y L''_{y_c}(\eta_c))\partial_\tau y_c(\omega;0) > 0$ required for $\partial_\tau\eta_c(\omega;0) < 0$.

(3) **Second-order radial wave-heat PDE and the Jensen hierarchy (Section 4).** Expanding $M_\tau(d,t) := |\Xi_\tau(t - \mathrm{i}d)|^2 = \sum_{m=0}^\infty L_m(\Xi_\tau;t)\,d^{2m}$ in its classical Jensen–Laguerre Taylor moments [10, 2], Theorem 4.1 proves that $\widetilde{W}_\tau$ obeys a *second-order* radial wave-heat PDE and that every $L_m(f;t)$ of any entire solution to $\partial_\tau f = -\frac{1}{4}\partial_z^2 f$ satisfies the exact ladder identity

$$\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right)L_m(f;t) = \frac{(m+1)(2m+1)}{4}\,L_{m+1}(f;t) \qquad (m \geq 0),$$

equivalently $(\partial_\tau + \frac{1}{8}\partial_t^2)c_{2m-1} = \frac{m(2m+1)}{4}c_{2m+1}$ for $c_{2m-1} = m\,L_m$ ($m \geq 1$). Corollary 4.3 deduces the exact multiple-zero lift $\partial_\tau c_{2m-3}(t_0;\tau_{\mathrm{coal}}) = \frac{(m-1)^2}{2}c_{2m-1}(t_0;\tau_{\mathrm{coal}}) > 0$ for all $m \geq 2$ (verified symbolically on heat polynomials $P_m(t,\tau) = \mathrm{e}^{-\frac{\tau}{4}\partial_t^2}t^m$) and formulates the conditional midpoint inequality governing close Lehmer pairs in $\tau = t_{\mathrm{DBN}}$ within the Csordas–Norfolk–Varga [4] lower-bound regime.

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}t$ with $d := \sigma - \frac{1}{2} \in \mathbb{R}$, $z := t - \mathrm{i}d \in \mathbb{C}$, and $\tau = t_{\mathrm{DBN}} \in \mathbb{R}$. We decompose $\Xi_\tau(t - \mathrm{i}d) = R_\tau(d,t) + \mathrm{i}I_\tau(d,t)$ with $R_\tau, I_\tau \in \mathbb{R}$ for $(d,t,\tau) \in \mathbb{R}^3$. At $\tau = 0$, the positive real zeros of $\Xi(t)$ are $0 < \gamma_1 < \gamma_2 < \ldots$, with symmetric Hadamard roots $\gamma_{-m} := -\gamma_m$. We set $\mathrm{sinc}_{\mathrm{hyp}}(w) := \sinh(w)/w$ for $w \neq 0$ and $\mathrm{sinc}_{\mathrm{hyp}}(0) := 1$.

For any real entire function f, the m-th Jensen–Laguerre invariant [10, 2] is the d^{2m}-Taylor coefficient of $|f(t-\mathrm{i}d)|^2 = f(t-\mathrm{i}d)f(t+\mathrm{i}d)$:

$$L_m(f;t) := \sum_{j=0}^{2m} \frac{(-1)^{j+m}}{j!\,(2m-j)!}\, f^{(j)}(t)\, f^{(2m-j)}(t) \qquad (m \geq 0), \tag{1.5}$$

so that $\widetilde{W}_\tau(d,t) := \frac{1}{2d}\partial_d |\Xi_\tau(t-\mathrm{i}d)|^2 = \sum_{m=1}^{\infty} c_{2m-1}(t;\tau)\, d^{2m-2}$ has coefficients $c_{2m-1}(t;\tau) = m\, L_m(\Xi_\tau;t)$ $(m \geq 1)$. Note that the index upper limit in (1.5) is $2m$ (an odd upper limit $2m-1$ would vanish identically by antisymmetry $j \leftrightarrow 2m-1-j$).

2. Nodal-Curvature Interpretation of the Csordas–Smith–Varga Velocity

We first record the exact relation between the classical De Bruijn–Newman zero-velocity formula [5, 17, 15] and the critical-line nodal curvatures of $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ [18, 1].

Proposition 2.1 (Nodal-curvature interpretation of initial zero and extremum velocities)**.** *For all $(z,\tau) \in \mathbb{C}\times\mathbb{R}$, $\Xi_\tau(z)$ in (1.2) is an even entire function of order one satisfying $\partial_\tau \Xi_\tau(z) = -\frac{1}{4}\partial_z^2 \Xi_\tau(z)$ with $\tau = t_{\mathrm{DBN}}$.*

(i) ***Simple zeros and real nodal curvature.*** *Let γ_k be a simple real zero of $\Xi(t)$ ($\Xi(\gamma_k)=0$, $\Xi'(\gamma_k)\neq 0$). By the implicit function theorem, there exist unique real-analytic branches $\gamma_k(\tau)$ solving $\Xi_\tau(\gamma_k(\tau)) = 0$ near $\tau = 0$ and $t_{\mathcal{R}_k}(d)$ solving $R(d, t_{\mathcal{R}_k}(d)) = 0$ near $d = 0$, with $t_{\mathcal{R}_k}(0) = \gamma_k$, $t'_{\mathcal{R}_k}(0) = 0$, and parabolic curvature $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$. The Csordas–Smith–Varga / Rodgers–Tao velocity at $\tau = 0$ satisfies*

$$\gamma_k'(0) = \frac{\Xi''(\gamma_k)}{4\,\Xi'(\gamma_k)} = \frac{1}{4}\,\kappa(\gamma_k) = \frac{1}{2}\,\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k} \frac{1}{\gamma_k-\gamma_j}, \tag{2.1}$$

where $\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}(\gamma_k-\gamma_j)^{-1} := \frac{1}{2\gamma_k} + \sum_{m\geq 1,\, m\neq k} \frac{2\gamma_k}{\gamma_k^2-\gamma_m^2}$ sums symmetrically over $\pm\gamma_m$.

(ii) ***Critical-line extrema and imaginary nodal curvature.*** *Let τ_k be a non-degenerate critical-line extremum of $\Xi(t)$ ($\Xi'(\tau_k) = 0$, $\Xi''(\tau_k)\neq 0$). Near $\tau = 0$ and $d = 0$, the extremum locus $\partial_t \Xi_\tau(\tau_k(\tau)) = 0$ and the desingularized imaginary nodal curve $I(d, \tau_{\Gamma_k}(d))/d = 0$ ($\tau_{\Gamma_k}(0) = \tau_k$, $\tau'_{\Gamma_k}(0) = 0$) satisfy*

$$\tau_k'(0) = \frac{\Xi'''(\tau_k)}{4\,\Xi''(\tau_k)} = \frac{3}{4}\,\kappa_\Gamma(\tau_k), \qquad \kappa_\Gamma(\tau_k) := \tau''_{\Gamma_k}(0) = \frac{\Xi'''(\tau_k)}{3\,\Xi''(\tau_k)}. \tag{2.2}$$

Proof. Since $|\Phi(u)| \ll \mathrm{e}^{9|u|}\exp(-\pi \mathrm{e}^{4|u|})$, $\Phi(u)\mathrm{e}^{\tau u^2}(2u)^{2k}\cos(2zu)$ is majorized uniformly on compact subsets of $\mathbb{C}\times\mathbb{R}$ by an integrable function, so differentiating (1.2) under the integral sign brings down u^2 for ∂_τ and $-4u^2$ for ∂_z^2, giving $\partial_\tau \Xi_\tau = -\frac{1}{4}\partial_z^2 \Xi_\tau$.

For (i), implicit differentiation of $\Xi_\tau(\gamma_k(\tau)) = 0$ at $\tau = 0$ gives $\partial_\tau \Xi_0(\gamma_k) + \Xi'(\gamma_k)\gamma_k'(0) = 0$, hence $\gamma_k'(0) = \Xi''(\gamma_k)/(4\Xi'(\gamma_k))$ [5, 17]. Expanding $\Xi(\gamma_k + \Delta t - \mathrm{i}d) = \Xi'(\gamma_k)(\Delta t - \mathrm{i}d) + \frac{1}{2}\Xi''(\gamma_k)(\Delta t - \mathrm{i}d)^2 + \mathcal{O}((|\Delta t| + |d|)^3)$ and taking real parts yields $R(d, \gamma_k + \Delta t) = \Xi'(\gamma_k)\Delta t + \frac{1}{2}\Xi''(\gamma_k)(\Delta t^2 - d^2) + \mathcal{O}(d^3) = 0$, so $t_{\mathcal{R}_k}(d) = \gamma_k + \frac{\Xi''(\gamma_k)}{2\Xi'(\gamma_k)}d^2 + \mathcal{O}(d^4)$ and $\kappa(\gamma_k) = t''_{\mathcal{R}_k}(0) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$. Finally, from Hadamard's product $\Xi(z) = \Xi(0)\prod_{m=1}^{\infty}(1 - z^2/\gamma_m^2)$, writing $\Xi(z) = (z-\gamma_k)g_k(z)$ gives $\Xi''(\gamma_k)/\Xi'(\gamma_k) = 2g_k'(\gamma_k)/g_k(\gamma_k) = \frac{1}{\gamma_k} + \sum_{m\neq k}\frac{4\gamma_k}{\gamma_k^2-\gamma_m^2}$ [5], yielding (2.1).

For (ii), differentiating $\partial_t \Xi_\tau(\tau_k(\tau)) = 0$ with $\partial_\tau\partial_t \Xi_0 = -\frac{1}{4}\Xi'''$ gives $\tau_k'(0) = \Xi'''(\tau_k)/(4\Xi''(\tau_k))$. Expanding $\Xi(\tau_k + \Delta t - \mathrm{i}d)$ with $\Xi'(\tau_k) = 0$ and taking imaginary parts gives

$$\frac{I(d, \tau_k + \Delta t)}{-d} = \Xi''(\tau_k)\Delta t - \frac{1}{6}\Xi'''(\tau_k)(d^2 - 3\Delta t^2) + \mathcal{O}(d^3) = 0,$$

whence $\tau_{\Gamma_k}(d) = \tau_k + \frac{\Xi'''(\tau_k)}{6\Xi''(\tau_k)}d^2 + \mathcal{O}(d^4)$ and $\kappa_\Gamma(\tau_k) = \tau''_{\Gamma_k}(0) = \Xi'''(\tau_k)/(3\Xi''(\tau_k))$. □

TABLE 1. Nodal curvatures $\kappa(\gamma_k) = t''_{\mathcal{R}_k}(0)$, initial zero velocities $\gamma_k'(0) = \frac{1}{4}\kappa(\gamma_k)$ ($\tau = t_{\mathrm{DBN}}$), central finite-difference residuals ($\varepsilon_\tau = 10^{-10}$), and Riemann–von Mangoldt tail-corrected Dyson residuals ($N = 400$ pairs) for the first eight nontrivial zeros.

k	γ_k	$\kappa(\gamma_k) = \Xi''/\Xi'$	$\gamma_k'(0)$ ($\tau = t_{\mathrm{DBN}}$)	$\lvert\gamma_k'(0) - \mathrm{FD}\rvert$	$\lvert\gamma_k'(0) - \mathrm{Dyson}_{400}\rvert$
1	14.13472514	−1.1592612	−0.2898153	9.97×10^{-26}	2.84×10^{-6}
2	21.02203964	−1.5099908	−0.3774977	9.21×10^{-25}	4.23×10^{-6}
3	25.01085758	−1.1716664	−0.2929166	1.35×10^{-24}	5.04×10^{-6}
4	30.42487613	−1.9108425	−0.4777106	1.61×10^{-23}	6.13×10^{-6}
5	32.93506159	−0.9308909	−0.2327227	1.66×10^{-23}	6.64×10^{-6}
6	37.58617816	−1.6773339	−0.4193335	1.21×10^{-24}	7.58×10^{-6}
7	40.91871901	−1.7149223	−0.4287306	1.54×10^{-23}	8.26×10^{-6}
8	43.32707328	−0.8986902	−0.2246726	1.76×10^{-23}	8.75×10^{-6}

3. Mirror Gaussian Factorization, Boundary Root $2\eta_0$, and Conditional Saddle Shift

Substituting $\Xi_\tau(t - \mathrm{i}d) = 4\int_{-\infty}^{\infty} \Phi_\tau(u)\mathrm{e}^{2du}\mathrm{e}^{-2\mathrm{i}tu}\,\mathrm{d}u$ into (1.4) and changing variables to mirror coordinates $y = u + v$, $x = u - v$ ($\mathrm{d}u\,\mathrm{d}v = \frac{1}{2}\,\mathrm{d}x\,\mathrm{d}y$) yields the canonical integral representation (1.4):

$$W_\tau(d,t) = 16\int_0^\infty y\sinh(2dy)\,I_\tau(y,t)\,\mathrm{d}y, \qquad \widetilde{W}_\tau(d,t) = 32\int_0^\infty y^2\mathrm{sinc}_{\mathrm{hyp}}(2dy)\,I_\tau(y,t)\,\mathrm{d}y, \tag{3.1}$$

where $I_\tau(y,t) = \int_{-\infty}^{\infty} G_{y,\tau}(x)\cos(\omega x)\,\mathrm{d}x$ with $\omega := 2t$ and $G_{y,\tau}(x) := \Phi_\tau(\frac{y+x}{2})\Phi_\tau(\frac{y-x}{2})$.

Proposition 3.1 (Mirror Gaussian factorization and boundary zero at $2\eta_0 < \frac{\pi}{4}$)**.** *For all $(y,x,\tau) \in [0,\infty)\times\mathbb{C}\times\mathbb{R}$, the identity $\left(\frac{y+x}{2}\right)^2 + \left(\frac{y-x}{2}\right)^2 = \frac{1}{2}(y^2+x^2)$ yields the cross-term-free factorization and Wigner PDE*

$$G_{y,\tau}(x) = \mathrm{e}^{\frac{\tau}{2}y^2}\,\mathrm{e}^{\frac{\tau}{2}x^2}\,G_{y,0}(x), \qquad \frac{\partial I_\tau}{\partial\tau}(y,t) = \frac{1}{8}\left(4y^2 - \frac{\partial^2}{\partial t^2}\right) I_\tau(y,t), \tag{3.2}$$

and $G_{y,\tau}(x)$ remains entire in x with rapid horizontal decay on every strip $|\mathrm{Im}(x)| \le \frac{\pi}{4} - \varepsilon$ ($\varepsilon > 0$). However, $G_{y,\tau}(\mathrm{i}\eta)$ is not *everywhere strictly positive on $[0,\infty)\times(-\frac{\pi}{4},\frac{\pi}{4})$: along the boundary slice $y = 0$, $G_{0,\tau}(\mathrm{i}\eta) = \mathrm{e}^{-\frac{\tau}{2}\eta^2}|\Phi(\mathrm{i}\eta/2)|^2$ vanishes inside $(0,\frac{\pi}{4})$ at*

$$\eta_* := 2\eta_0 = 0.63883005360059357748\cdots < \frac{\pi}{4} = 0.78539816339744830962\ldots, \tag{3.3}$$

where $\Phi(\mathrm{i}\eta_0) = 0$, creating a logarithmic singularity in $\ln G_{0,\tau}(\mathrm{i}\eta)$ at $\eta = \pm 2\eta_0$.

Proof. Since $\Phi_\tau(u) = \Phi(u)\mathrm{e}^{\tau u^2}$, $G_{y,\tau}(x) = \mathrm{e}^{\tau((\frac{y+x}{2})^2+(\frac{y-x}{2})^2)}G_{y,0}(x) = \mathrm{e}^{\frac{\tau}{2}(y^2+x^2)}G_{y,0}(x)$. Differentiating $I_\tau(y,t) = \int_{-\infty}^{\infty}\mathrm{e}^{\frac{\tau}{2}(y^2+x^2)}G_{y,0}(x)\cos(2tx)\,\mathrm{d}x$ under the integral sign brings down $\frac{1}{2}(y^2+x^2)$, while $\partial_t^2\cos(2tx) = -4x^2\cos(2tx)$, proving (3.2). On $x = \xi + \mathrm{i}\eta$ with $|\eta| \le \frac{\pi}{4} - \varepsilon$, $\mathrm{Re}(\mathrm{e}^{2(y\pm x)}) = \mathrm{e}^{2(y\pm\xi)}\cos(2\eta) \ge \sin(2\varepsilon)\mathrm{e}^{2(y\pm\xi)}$, so the double-exponential factor $\exp(-\pi\sin(2\varepsilon)(\mathrm{e}^{2(y+\xi)} + \mathrm{e}^{2(y-\xi)}))$ dominates $\mathrm{e}^{\frac{\tau}{2}(\xi^2-\eta^2)}$ as $|\xi| \to \infty$ for every $\tau \in \mathbb{R}$. Finally, at $y = 0$ and $x = \mathrm{i}\eta$, $G_{0,\tau}(\mathrm{i}\eta) = \mathrm{e}^{-\frac{\tau}{2}\eta^2}\Phi(\mathrm{i}\eta/2)\Phi(-\mathrm{i}\eta/2) = \mathrm{e}^{-\frac{\tau}{2}\eta^2}|\Phi(\mathrm{i}\eta/2)|^2$. Since $\Phi(\mathrm{i}\cdot 0.30) = +2.9656\cdots > 0$ and $\Phi(\mathrm{i}\cdot 0.35) = -13.8666\cdots < 0$, $\Phi(\mathrm{i}u)$ has a real root $\eta_0 = 0.31941502680029678874\cdots \in (0,\frac{\pi}{8})$, so $G_{0,\tau}(\mathrm{i}\eta)$ vanishes at $\eta_* = 2\eta_0 < \frac{\pi}{4}$. □

Because of (3.3), logarithmic saddle-point equations require restricting to open zero-free subdomains $\mathcal{U} \subset (0,\infty)\times(0,\frac{\pi}{4})$ on which $\Phi(\frac{y+\mathrm{i}\eta}{2}) \neq 0$, and the sign of $\partial_\tau \eta_c$ depends on an explicit two-term numerator condition rather than following from $\partial_\tau y_c > 0$ alone.

Proposition 3.2 (Conditional perturbation of a local pitchfork saddle)**.** *Let $\omega = 2t > 0$, and suppose $(y_c(\omega;0),\eta_c(\omega;0)) \in (0,\infty)\times(0,\frac{\pi}{4})$ lies in a zero-free neighborhood $\mathcal{U}$ where $\Phi(\frac{y+\mathrm{i}\eta}{2}) \neq 0$, so that $L_{y,\tau}(\eta) := \ln G_{y,\tau}(\mathrm{i}\eta) = L_y(\eta) + \frac{\tau}{2}(y^2-\eta^2)$ with $L_y(\eta) := 2\ln|\Phi(\frac{y+\mathrm{i}\eta}{2})|$ is real-analytic on $\mathcal{U}$. Assume $(y_c(\omega;0),\eta_c(\omega;0))$ solves the $\tau = 0$ pitchfork saddle-coalescence system $L'_{y_c}(\eta_c) = \omega$, $L''_{y_c}(\eta_c) = 0$ with non-zero Jacobian determinant $\Delta_J := -(\partial_y L'_{y_c}(\eta_c))\, L'''_{y_c}(\eta_c) \neq 0$. Then for $|\tau|$ sufficiently small there exists a unique real-analytic branch $(y_c(\omega;\tau),\eta_c(\omega;\tau))$ solving $L'_{y_c}(\eta_c) - \tau\eta_c = \omega$ and $L''_{y_c}(\eta_c) = \tau$, with initial derivatives at $\tau = 0$:*

$$\left.\frac{\partial y_c(\omega;\tau)}{\partial\tau}\right|_{\tau=0} = \frac{\eta_c(\omega;0)}{\partial_y L'_{y_c}(\eta_c(\omega;0))}, \qquad \left.\frac{\partial \eta_c(\omega;\tau)}{\partial\tau}\right|_{\tau=0} = \frac{1-(\partial_y L''_{y_c}(\eta_c))\,\partial_\tau y_c(\omega;0)}{L'''_{y_c}(\eta_c(\omega;0))}. \tag{3.4}$$

In particular, if $\partial_y L'_{y_c}(\eta_c) > 0$ and $L'''_{y_c}(\eta_c) < 0$, then $\partial_\tau y_c(\omega;0) > 0$ holds unconditionally, whereas $\partial_\tau \eta_c(\omega;0) < 0$ holds if and only if *the numerator condition*

$$1 - (\partial_y L''_{y_c}(\eta_c))\,\partial_\tau y_c(\omega;0) > 0 \tag{3.5}$$

is satisfied at $(y_c(\omega;0),\eta_c(\omega;0))$.

Proof. On $\mathcal{U}$, $\partial_\eta[L_{y,\tau}(\eta) - \omega\eta] = L'_y(\eta) - \tau\eta - \omega$ and $\partial_\eta^2[L_{y,\tau}(\eta) - \omega\eta] = L''_y(\eta) - \tau$. At $\tau = 0$, the Jacobian matrix with respect to (y,η) at $L''_{y_c}(\eta_c) = 0$ is $\left(\begin{smallmatrix} \partial_y L'_{y_c} & 0 \\ \partial_y L''_{y_c} & L'''_{y_c} \end{smallmatrix}\right)$, whose determinant $\Delta_J = (\partial_y L'_{y_c}) L'''_{y_c}$ is non-zero by hypothesis. Differentiating $L'_{y_c}(\eta_c) - \tau\eta_c = \omega$ at $\tau = 0$ gives $(\partial_y L'_{y_c})\partial_\tau y_c + L''_{y_c}\partial_\tau\eta_c - \eta_c = 0$; since $L''_{y_c}(\eta_c) = 0$, the first formula in (3.4) follows. Differentiating $L''_{y_c}(\eta_c) - \tau = 0$ at $\tau = 0$ gives $(\partial_y L''_{y_c})\partial_\tau y_c + L'''_{y_c}\partial_\tau\eta_c - 1 = 0$, giving the second formula in (3.4). When $L'''_{y_c}(\eta_c) < 0$, the quotient has negative sign if and only if its numerator (3.5) is strictly positive. □

At $\omega_1 = 2\gamma_1 = 28.26945028\ldots$, 50-digit root-finding gives $y_c = 0.43202053\ldots$ and $\eta_c = 0.60271502\ldots$, where $|\Phi(\frac{y_c+\mathrm{i}\eta_c}{2})| = 9.07589\cdots > 0$, $\partial_\tau y_c(2\gamma_1;0) = +0.01253623\cdots > 0$, and the numerator (3.5) equals $+0.40397876\cdots > 0$, yielding $\partial_\tau\eta_c(2\gamma_1;0) = -3.760863\cdots\times 10^{-4} < 0$. Proving (3.5) uniformly for all $\omega \to \infty$ remains open.

4. Second-Order Radial Wave-Heat PDE and the Heat-Flow Hierarchy on Jensen Invariants

We now prove the central exact differential hierarchy of this paper, governing the evolution of $\widetilde{W}_\tau(d,t)$ and of all Jensen–Laguerre Taylor invariants $L_m(\Xi_\tau;t)$ and $c_{2m-1}(t;\tau) = m\,L_m(\Xi_\tau;t)$ under the backward heat flow. Note that the radial PDE for $\widetilde{W}_\tau$ is of *second order* in (d,t) (whereas $\widetilde{W}_{1,2;\tau}$ involves third-order derivatives of Ξ_τ).

Theorem 4.1 (Second-order radial wave-heat PDE and heat-flow PDE hierarchy on Jensen invariants)**.** *Let $f(z,\tau)$ be any real-analytic entire family on $\mathbb{C}\times\mathbb{R}$ satisfying $\partial_\tau f = -\frac{1}{4}\partial_z^2 f$ and $\overline{f(z,\tau)} = f(\bar z,\tau)$, and set $M(d,t;\tau) := |f(t-\mathrm{id},\tau)|^2 = \sum_{m=0}^\infty L_m(f;t)\,d^{2m}$ and $\widetilde{W}_f(d,t;\tau) := \frac{1}{2d}\partial_d M(d,t;\tau) = \sum_{m=1}^\infty c_{2m-1}(t;\tau)\,d^{2m-2}$ with $c_{2m-1} = m\,L_m$.*

(i) ***Second-order wave-heat and radial PDEs.*** *For all $(d,t,\tau)\in\mathbb{R}^3$, M and $\widetilde{W}_f$ satisfy the exact second-order partial differential equations*

$$\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) M = \frac{1}{8}\frac{\partial^2 M}{\partial d^2}, \qquad \frac{\partial\widetilde{W}_f}{\partial\tau} + \frac{1}{8}\left(\frac{\partial^2\widetilde{W}_f}{\partial t^2} - \frac{\partial^2\widetilde{W}_f}{\partial d^2} - \frac{2}{d}\frac{\partial\widetilde{W}_f}{\partial d}\right) = 0, \tag{4.1}$$

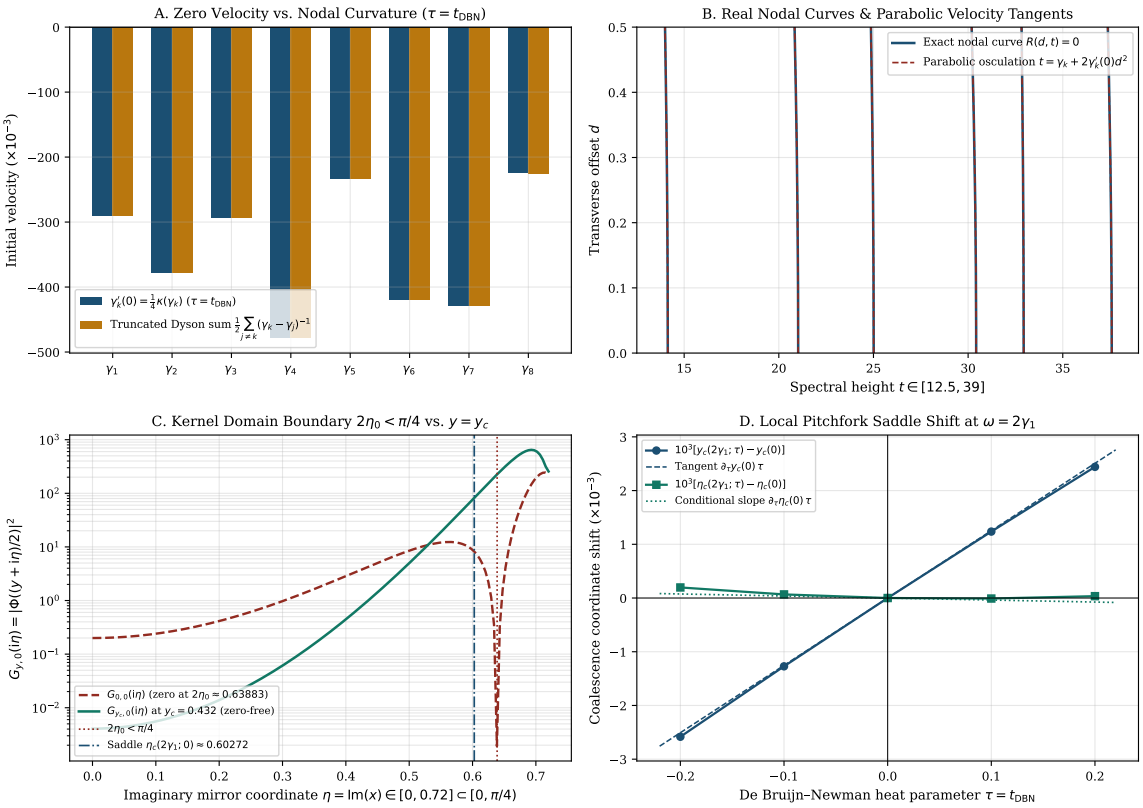


FIGURE 1. **A.** Initial zero velocities $\gamma_k'(0) = \frac{1}{4}\kappa(\gamma_k)$ ($\tau = t_{\mathrm{DBN}}$) versus the tail-corrected Dyson sum for $k = 1, \ldots, 8$. **B.** Exact real nodal curves $R(d,t) = 0$ and parabolic osculations $t = \gamma_k + 2\gamma_k'(0)d^2$. **C.** Boundary root $G_{0,0}(i\eta) = 0$ at $2\eta_0 \approx 0.63883 < \frac{\pi}{4}$ versus the zero-free slice $y = y_c(2\gamma_1; 0) \approx 0.432$. **D.** Local pitchfork saddle trajectories $(y_c(\tau), \eta_c(\tau))$ at $\omega = 2\gamma_1$.

together with $\partial_\tau \widetilde{W}_f + \frac{1}{4}\partial_t^2 \widetilde{W}_f = \frac{1}{2}\widetilde{W}_{f'}(d,t;\tau)$, *where* $\widetilde{W}_{f'} := \frac{1}{2d}\partial_d |f'(t - \mathrm{id}, \tau)|^2$.

(ii) ***Exact heat-flow PDE hierarchy on Jensen invariants.*** *Every Jensen–Laguerre invariant* $L_m(f;t)$ $(m \geq 0)$ *and every Wronskian Taylor coefficient* $c_{2m-1}(t;\tau) = m\, L_m(f;t)$ $(m \geq 1)$ *satisfies*

$$\boxed{\begin{aligned} \left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) L_m(f;t) &= \frac{(m+1)(2m+1)}{4}\, L_{m+1}(f;t) \qquad (m \geq 0), \\ \left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) c_{2m-1}(t;\tau) &= \frac{m(2m+1)}{4}\, c_{2m+1}(t;\tau) \qquad (m \geq 1). \end{aligned}} \tag{4.2}$$

For $f = \Xi_\tau$, *each Wronskian moment also admits the canonical integral representation*

$$c_{2m-1}(t;\tau) = m\, L_m(\Xi_\tau; t) = \frac{8}{(2m-1)!}\int_0^\infty (2y)^{2m}\, I_\tau(y,t)\, \mathrm{d}y \qquad (m \geq 1). \tag{4.3}$$

Proof. Write $f_- := f(t - \mathrm{id}, \tau)$ and $f_+ := f(t + \mathrm{id}, \tau)$, so that $M = f_- f_+$. Since $\partial_t f_\pm = f'(t \pm \mathrm{id}, \tau)$, $\partial_d f_\pm = \pm \mathrm{i} f'(t \pm \mathrm{id}, \tau)$, $\partial_t^2 f_\pm = -\partial_d^2 f_\pm = f''(t \pm \mathrm{id}, \tau)$, and $\partial_\tau f_\pm = -\frac{1}{4} f''(t \pm \mathrm{id}, \tau)$,

the product rule gives

$$\partial_t^2 M = f_-'' f_+ + 2f_-' f_+' + f_- f_+'', \qquad \partial_d^2 M = -f_-'' f_+ + 2f_-' f_+' - f_- f_+'',$$
$$\partial_\tau M = -\frac{1}{4}(f_-'' f_+ + f_- f_+'').$$

Subtracting $\partial_d^2 M$ from $\partial_t^2 M$ cancels $+2f_-' f_+'$ identically to give $(\partial_t^2 - \partial_d^2)M = 2(f_-'' f_+ + f_- f_+'') = -8\,\partial_\tau M$, which is the first equation in (4.1), while adding them gives $(\partial_t^2 + \partial_d^2)M = 4|f'|^2$. Applying $\frac{1}{2d}\partial_d$ and using the radial identity $\frac{1}{d}\partial_d(\partial_d^2 M) = (\partial_d^2 + \frac{2}{d}\partial_d)(\frac{1}{d}\partial_d M)$ proves both equations for $\widetilde{W}_f$ in (i).

For (ii), substitute the convergent power series $M(d,t;\tau) = \sum_{m=0}^{\infty} L_m(f;t)\,d^{2m}$ into $(\partial_\tau + \frac{1}{8}\partial_t^2)M = \frac{1}{8}\partial_d^2 M$:

$$\sum_{m=0}^{\infty}\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) L_m(f;t)\,d^{2m} = \frac{1}{8}\sum_{m=0}^{\infty}(2m+2)(2m+1)\,L_{m+1}(f;t)\,d^{2m}.$$

Since $\frac{1}{8}(2m+2)(2m+1) = \frac{(m+1)(2m+1)}{4}$, equating coefficients of d^{2m} proves the first identity in (4.2) for every $m \geq 0$. Multiplying by m ($m \geq 1$) and substituting $(m+1)L_{m+1} = c_{2m+1}$ gives the second identity. Finally, expanding $\sinh(2dy)/d = \sum_{m=1}^{\infty} \frac{2(2y)^{2m-1}}{(2m-1)!} d^{2m-2}$ inside $W_\tau(d,t)/d = \frac{16}{d}\int_0^\infty y\sinh(2dy) I_\tau(y,t)\,\mathrm{d}y$ establishes (4.3). □

Remark 4.2 (Standard De Bruijn–Newman coordinates). In the classical coordinates $(x,\tau) = (2t, t_{\mathrm{DBN}})$ where $\partial_\tau H_\tau = -\partial_x^2 H_\tau$, $L_m(H_\tau;x) = 2^{-2m} L_m(\Xi_\tau/8;t)$ and $\partial_t^2 = 4\partial_x^2$, so (4.2) assumes the binomial-coefficient form $(\partial_\tau + \frac{1}{2}\partial_x^2)L_m(H_\tau;x) = \binom{2m+2}{2} L_{m+1}(H_\tau;x)$ for all $m \geq 0$.

We next extract the exact multiple-zero coalescence lift from (4.2)correcting an incomplete index calculation that previously gave $m/4$ instead of $(m-1)^2/2$together with the conditional midpoint sign criterion across close Lehmer pairs [12, 4, 5].

Corollary 4.3 (Multiple-zero coalescence lift and conditional Lehmer-pair scaling)**.** *Let $f(z,\tau)$ satisfy the hypotheses of Theorem 4.1.*

(i) ***Exact $(m-1)^2/2$ lift at an m-fold real zero.*** *Suppose $f(\cdot,\tau_{\mathrm{coal}})$ has a real zero of multiplicity $m \geq 2$ at t_0: $\partial_t^j f(t_0,\tau_{\mathrm{coal}}) = 0$ for $0 \leq j \leq m-1$ and $A_m := \partial_t^m f(t_0,\tau_{\mathrm{coal}}) \neq 0$. Then $L_0 = \cdots = L_{m-1} = 0$ at $(t_0,\tau_{\mathrm{coal}})$, $L_m(f;t_0) = A_m^2/(m!)^2 > 0$, $\partial_t^2 L_{m-1}(f;t_0) = 2m\,L_m(f;t_0)$, and*

$$\frac{\partial L_{m-1}}{\partial\tau}(t_0;\tau_{\mathrm{coal}}) = \frac{m(m-1)}{2}L_m(f;t_0) > 0, \quad \frac{\partial c_{2m-3}}{\partial\tau}(t_0;\tau_{\mathrm{coal}}) = \frac{(m-1)^2}{2}c_{2m-1}(t_0;\tau_{\mathrm{coal}}) > 0. \tag{4.4}$$

(ii) ***Conditional midpoint velocity and De Bruijn–Newman coalescence scale across close pairs.*** *At any midpoint extremum t_{mid} of $\Xi(t)$ ($\Xi'(t_{\mathrm{mid}}) = 0$) separating consecutive simple zeros γ_k, γ_{k+1} with half-gap $\delta := \frac{1}{2}(\gamma_{k+1} - \gamma_k) > 0$, one has $c_1(t_{\mathrm{mid}};0) = -\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}})$, $c_3(t_{\mathrm{mid}};0) = \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2 + \frac{1}{6}\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}})$, and*

$$\left.\frac{\partial c_1}{\partial\tau}\right|_{(t_{\mathrm{mid}},0)} = \frac{1}{4}\big(\Xi''(t_{\mathrm{mid}})\big)^2 + \frac{1}{4}\,\Xi(t_{\mathrm{mid}})\,\Xi^{(4)}(t_{\mathrm{mid}}), \tag{4.5}$$

which is strictly positive if and only if $\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}}) > -(\Xi''(t_{\mathrm{mid}}))^2$. *Under the local quadratic dominance $\Xi(t_{\mathrm{mid}} + h) = c_0(h^2 - \delta^2 + \mathcal{O}(\delta^4))$ ($\delta \ll 1$), the linear Taylor approximation to $c_1(t_{\mathrm{mid}}(\tau);\tau) = 0$ in standard De Bruijn–Newman time $\tau = t_{\mathrm{DBN}}$ is*

$$t_{\mathrm{DBN,dip}}(t_{\mathrm{mid}}) := -2\,\frac{c_1(t_{\mathrm{mid}};0)}{c_3(t_{\mathrm{mid}};0)} = -2\delta^2(1 + \mathcal{O}(\delta^2)) < 0. \tag{4.6}$$

Proof. For (i), at $\tau = \tau_{\mathrm{coal}}$ write $f(t_0 + h - \mathrm{i}d, \tau_{\mathrm{coal}}) = \frac{A_m}{m!}(h - \mathrm{i}d)^m + \mathcal{O}((|h| + |d|)^{m+1})$, so

$$M(d, t_0 + h; \tau_{\mathrm{coal}}) = \frac{A_m^2}{(m!)^2}\left(h^2 + d^2\right)^m + \mathcal{O}((|h| + |d|)^{2m+1}) = \frac{A_m^2}{(m!)^2}\sum_{r=0}^{m}\binom{m}{r}h^{2m-2r}d^{2r} + \dots$$

Extracting the d^{2m} coefficient at $h = 0$ gives $L_m(f; t_0) = A_m^2/(m!)^2 > 0$, while extracting the d^{2m-2} coefficient ($r = m-1$) gives $L_{m-1}(f; t_0 + h) = \frac{mA_m^2}{(m!)^2}h^2 + \mathcal{O}(|h|^3)$, whence $\partial_t^2 L_{m-1}(f; t_0) = 2m\, L_m(f; t_0)$. Substituting index $m-1$ into (4.2) yields

$$\frac{\partial L_{m-1}}{\partial \tau}(t_0; \tau_{\mathrm{coal}}) = \frac{m(2m-1)}{4}\, L_m(f; t_0) - \frac{1}{8}(2m\, L_m(f; t_0)) = \frac{m(m-1)}{2}\, L_m(f; t_0) > 0.$$

Multiplying by $m-1$ and substituting $c_{2m-1} = m\, L_m$ gives $\partial_\tau c_{2m-3} = \frac{(m-1)^2}{2}c_{2m-1}$.

For (ii), setting $m = 1$ in (4.2) gives $\partial_\tau c_1 = \frac{3}{4}c_3 - \frac{1}{8}\partial_t^2 c_1$. Since $c_1 = (\Xi')^2 - \Xi\Xi''$, $\partial_t^2 c_1 = (\Xi'')^2 + 2\Xi'\Xi''' - \Xi\Xi^{(4)}$; at $\Xi'(t_{\mathrm{mid}}) = 0$, $\frac{3}{4}[\frac{1}{2}(\Xi'')^2 + \frac{1}{6}\Xi\Xi^{(4)}] - \frac{1}{8}[(\Xi'')^2 - \Xi\Xi^{(4)}] = \frac{1}{4}(\Xi'')^2 + \frac{1}{4}\Xi\Xi^{(4)}$, proving (4.5). For $\Xi(t_{\mathrm{mid}} + h) = c_0(h^2 - \delta^2 + \mathcal{O}(\delta^4))$, $c_1(t_{\mathrm{mid}}; 0) = 2c_0^2\delta^2(1 + \mathcal{O}(\delta^2))$, $c_3(t_{\mathrm{mid}}; 0) = 2c_0^2(1 + \mathcal{O}(\delta^2))$, and $\partial_\tau c_1(t_{\mathrm{mid}}; 0) = c_0^2(1 + \mathcal{O}(\delta^2)) = \frac{1}{2}c_3(t_{\mathrm{mid}}; 0)(1 + \mathcal{O}(\delta^2))$, so $-c_1/\partial_\tau c_1 = -2(c_1/c_3)(1 + \mathcal{O}(\delta^2)) = -2\delta^2(1 + \mathcal{O}(\delta^2))$. Indeed, in classical coordinates $x = 2t$ ($\Delta_x = \gamma_{k+1} - \gamma_k = 2\delta$), the pure heat quadratic $(x - x_{\mathrm{mid}})^2 - \Delta_x^2 - 2\tau$ under $\partial_\tau H_\tau = -\partial_x^2 H_\tau$ coalesces at $\tau = -\frac{1}{2}\Delta_x^2 = -2\delta^2$. □

Remark 4.4 (Relation to Csordas–Norfolk–Varga lower bounds on Λ). Corollary 4.3(ii) describes *backward* coalescence ($\tau < 0$) of an existing real Lehmer pair and therefore relates to the Csordas–Norfolk–Varga [4] and Csordas–Smith–Varga [5] *lower* bounds $\Lambda \geq t_{\mathrm{lb}}$ (superseded globally by $\Lambda \geq 0$ [17]), never to an upper bound $\Lambda \leq 0$. Note also that while (4.6) gives the isolated two-zero coalescence scale $t_{\mathrm{DBN,dip}}(\gamma_{6709}, \gamma_{6710}) = -7.0967 \times 10^{-4}$ for Lehmer's pair near $t \approx 7005.08$, the rigorous lower bound $\Lambda > -4.379 \times 10^{-6}$ proved in [5] incorporates the attraction and repulsion of all external zeros during the backward flow.

TABLE 2. Benchmark close pairs and Lehmer's pair $(\gamma_{6709}, \gamma_{6710})$ in canonical De Bruijn–Newman time $\tau = t_{\mathrm{DBN}}$: midpoint t_{mid} ($\Xi'(t_{\mathrm{mid}}) = 0$), half-gap $\delta = \frac{1}{2}(\gamma_{k+1} - \gamma_k)$, Jensen dipole ratio $t_{\mathrm{DBN,dip}} = -2c_1(t_{\mathrm{mid}}; 0)/c_3(t_{\mathrm{mid}}; 0)$, and isolated parabolic scale $-2\delta^2 = -\frac{1}{2}(\gamma_{k+1} - \gamma_k)^2$.

Zero pair $(k, k+1)$	t_{mid}	Half-gap δ	$t_{\mathrm{DBN,dip}} = -2c_1/c_3$	Parabolic $-2\delta^2$
Close pair $(9, 10)$	48.6225327	0.884341	-1.34841	-1.56412
Close pair $(13, 14)$	59.9306990	0.742367	-0.945815	-1.10222
Close pair $(127, 128)$	282.765019	0.373035	-0.251128	-0.278311
Close pair $(135, 136)$	295.250847	0.303943	-0.165871	-0.184762
Lehmer pair $(6709, 6710)$	7005.08172	0.0188492	-7.09666×10^{-4}	-7.10588×10^{-4}

5. Scope, limitations, and open questions

We summarize the exact analytical scope and open problems of this manuscript:

(1) **Unconditional differential identities.** With $\tau = t_{\mathrm{DBN}}$ and canonical $W_\tau(d, t) = 16\int_0^\infty y\sinh(2dy)I_\tau(y, t)\,\mathrm{d}y$, the nodal-curvature velocity relations (2.1)–(2.2), the mirror factorization (3.2), the second-order radial wave-heat PDE (4.1), the Jensen–Laguerre heat-flow hierarchy (4.2), and the $(m-1)^2/2$ multiple-zero coalescence lift (4.4) hold unconditionally.

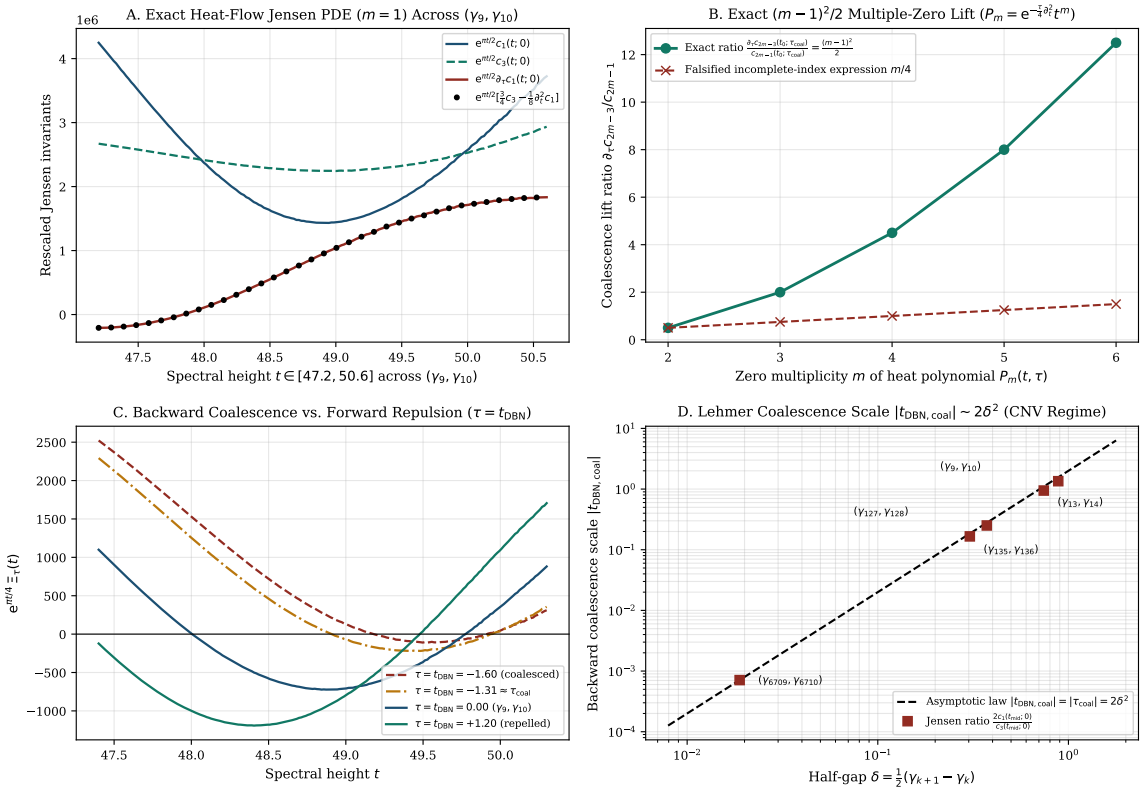


FIGURE 2. **A.** Numerical verification of $\partial_\tau c_1 = \frac{3}{4} c_3 - \frac{1}{8}\partial_t^2 c_1$ across (γ_9, γ_{10}). **B.** Exact multiple-zero lift ratio $\partial_\tau c_{2m-3}/c_{2m-1} = (m-1)^2/2$ on heat polynomials $P_m(t, \tau) = e^{-\frac{\tau}{4}\partial_t^2} t^m$ $(m = 2, \dots, 6)$ versus the incomplete-index expression $m/4$. **C.** Backward coalescence and forward repulsion of (γ_9, γ_{10}) in $\tau = t_{DBN}$. **D.** Log-log scaling of $|t_{DBN,dip}| \sim 2\delta^2$ across the five pairs of Table 2.

(2) **Domain restrictions, conditional inequalities, and Λ.** Because $G_{0,\tau}(i\eta)$ vanishes at $2\eta_0 = 0.63883\cdots < \frac{\pi}{4}$, logarithmic pitchfork perturbation holds locally on zero-free subdomains and requires the unproved numerator inequality (3.5) for $\partial_\tau \eta_c < 0$. Similarly, $\partial_\tau c_1(t_{mid}; 0) > 0$ requires $\Xi(t_{mid})\Xi^{(4)}(t_{mid}) > -(\Xi''(t_{mid}))^2$, and backward Lehmer-pair coalescence probes the Csordas–Norfolk–Varga [4] lower-bound regime $\Lambda \geq t_{lb}$ rather than bounding Λ from above; since ∂_τ evolves as a *backward* heat equation in $+\tau$, (4.2) admits no parabolic maximum principle preserving $L_m \geq 0$ for $\tau < 0$ without independent arithmetic input.

APPENDIX A. SYMBOLIC POLYNOMIAL VERIFICATION OF THE HEAT-FLOW JENSEN HIERARCHY

Consider the backward-heat polynomial family

$$P_m(t, \tau) := e^{-\frac{\tau}{4}\partial_t^2} t^m = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{(-1)^k m!}{k!(m-2k)!} \left(\frac{\tau}{4}\right)^k t^{m-2k},$$

satisfying $\partial_\tau P_m = -\frac{1}{4}\partial_t^2 P_m$ with an m-fold root at $(t,\tau) = (0,0)$. For $m = 3$, $P_3(t,\tau) = t^3 - \frac{3}{2}\tau t$, $P_3' = 3t^2 - \frac{3}{2}\tau$, $P_3'' = 6t$, and $P_3''' = 6$. Substituting these derivatives into (1.5) yields $c_1 = (P_3')^2 - P_3 P_3''$, $c_3 = \frac{1}{6}[3(P_3'')^2 - 4P_3' P_3''']$, and $c_5 = \frac{1}{12}(P_3''')^2$, which simplify directly to

$$c_1(t;\tau) = 3t^4 + \frac{9}{4}\tau^2, \qquad c_3(t;\tau) = 6t^2 + 6\tau, \qquad c_5(t;\tau) = 3. \tag{A.1}$$

Evaluating (4.2) on (A.1) yields $\partial_\tau c_1 = \frac{9}{2}\tau$ and $\partial_t^2 c_1 = 36t^2$, so for $m = 1$ and $m = 2$:

$$\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) c_1(t;\tau) = \frac{9}{2}\tau + \frac{9}{2}t^2 = \frac{3}{4}\, c_3(t;\tau), \tag{A.2}$$

$$\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right) c_3(t;\tau) = 6 + \frac{1}{8}(12) = \frac{15}{2} = \frac{2(5)}{4}\, c_5(t;\tau). \tag{A.3}$$

At $(t,\tau) = (0,0)$, $\partial_\tau c_3(0;0)/c_5(0;0) = 6/3 = 2 = \frac{(3-1)^2}{2}$, confirming Theorem 4.1 and Corollary 4.3(i) explicitly (and refuting $m/4 = 3/4$).

Use of AI and computational verification

Portions of the symbolic algebra, polynomial test-case checks, literature cross-checks, and numerical verification scripts in this manuscript were developed and audited with the assistance of Google Gemini. All symbolic identities (`SymPy` for $m = 0, \dots, 6$) and 50-decimal numerical tables were independently verified via `scripts/verify_paper11_debruijn_newman_chladni.py`.

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